

The Problem & Its Real Ceiling

- Classify public statements as *True/False* from text + metadata (speaker, party, subjects, context)
- This is a low-ceiling task: **~62–69% binary accuracy** across model families and methods [1]
- **Why:** verifying claims needs external world knowledge the text lacks; human truth labels are judgment calls with limited agreement; some statements aren't verifiable from inputs
- **Aim:** a principled LLM-based evaluation pipeline

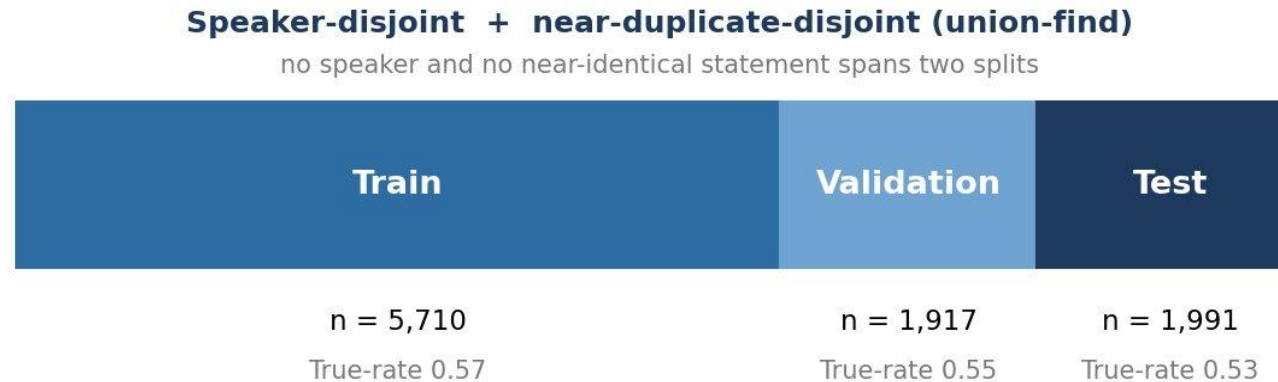
Mapping the 6 Labels To Binary Classification

- Source: 6 ordinal truthfulness levels
- **Middle split:** 3 truer $\rightarrow$ True, 3 falser $\rightarrow$ False
- **Why:** near-balanced target ($\approx 56/44$) keeps accuracy meaningful, no majority-class
- **Sensitivity check:** moving ambiguous "half-true" to False leaves conclusions unchanged

Data Integrity: Split That Prevents Cheating

- **Naive random splits leak:** frequent speakers appear in train and test → model learns "this person lies" not the claim

- **My split** is the following:



- Dropped contradictory duplicates (identical text, conflicting labels)
- Matched class balance across splits, thus, the train/test comparison isn't distorted by class drift

Evaluation Methodology: Metrics & Protocol

- Beyond simple *accuracy*: *balanced accuracy*, *macro-F1*, *ROC/PR-AUC*, and *calibration (Brier + ECE)*
- Cost asymmetry: a false positive lets misinformation through and it is the costlier error; threshold can reflect that
- Protocol: calibrator + threshold fit on validation set, reported on untouched test. Bootstrap CIs; McNemar's test for the paired comparison

Zero-Shot Predictor

- Base: Gemma-4-31B-it (open; serverless on Together.ai; LoRA-fine-tunable under the same id → shared base, no confound)
- Prompt: statement + metadata, missing fields marked [unknown]
- Elicitation: decision + logprob — model answers True/False; confidence from the True/False token's *logprob* [2]
- Baseline (calibrated, test): acc 0.668, ECE 0.053

[2] <https://docs.together.ai/docs/inference/chat/logprobs>

Fine-Tuned Predictor (LoRA SFT)

- LoRA supervised fine-tuning on the same Gemma-4-31B-it model
- Training data: conversational JSONL
 - *prompt* as system+user, *assistant* = single True/False token
 - *loss* on the label only (train_on_inputs=auto)
- Train split only (5,710); validation file for eval-loss. Eval loss 0.32 → 0.199 over 3 epochs
- Serving: fine-tuned Gemma isn't serverless, thus, we use a short-lived dedicated endpoint

The Fair Comparison & Results

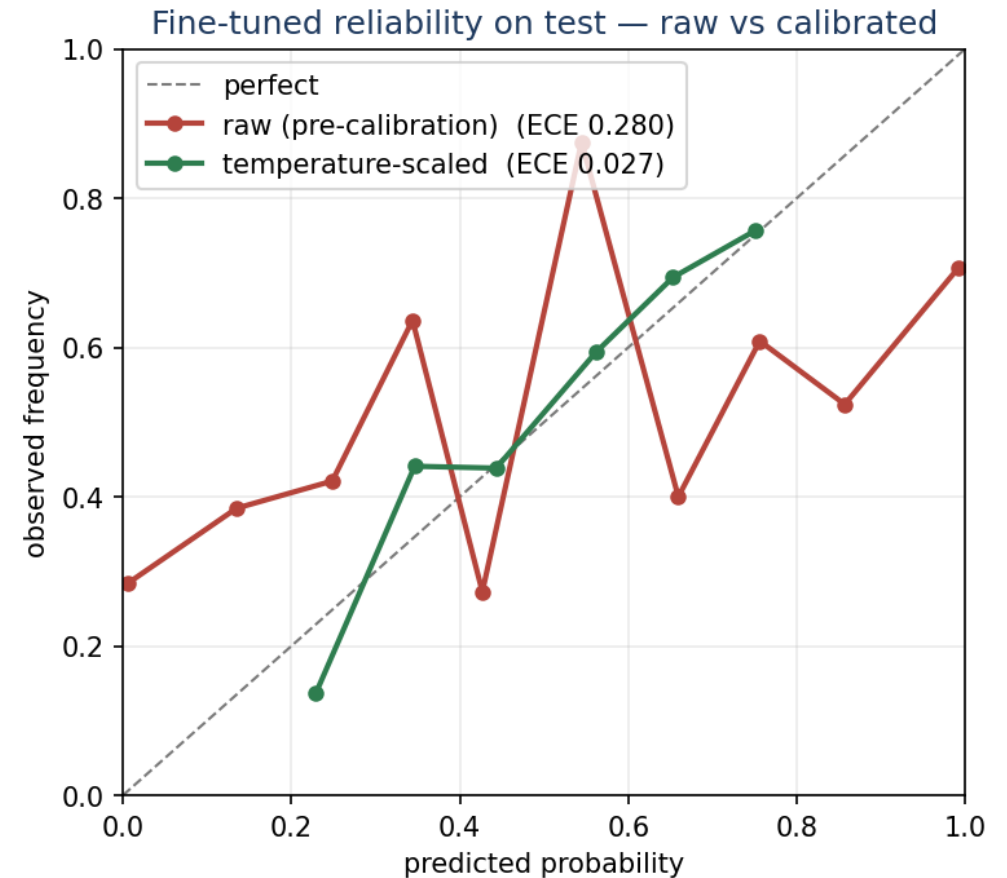
- Same test split, same calibration, same elicitation; only fine-tuning differs

Predictor	Acc	Bal-Acc	Macro-F1	ROC-AUC	PR-AUC	Brier	ECE
Zero-shot (decision/logprob)	<u>0.668</u>	<u>0.662</u>	<u>0.662</u>	<u>0.711</u>	<u>0.679</u>	<u>0.217</u>	<u>0.053</u>
Fine-tuned (LoRA SFT, decision)	0.701	0.695	0.696	0.749	0.716	0.201	0.027
Zero-shot (score)	0.700	0.699	0.699	0.715	0.660	0.225	0.117

- **Clean fine-tuning effect (matched elicitation):** $\Delta\text{acc} +0.033$ [$+0.021$, $+0.045$], McNemar $p < 0.001$. Calibration: ECE $0.053 \rightarrow 0.027$.
- **Secondary (score-mode):** highest accuracy yet $\sim 4\times$ worse calibrated (ECE 0.117 vs 0.027)
Therefore, ***accuracy $\neq$ trustworthiness***

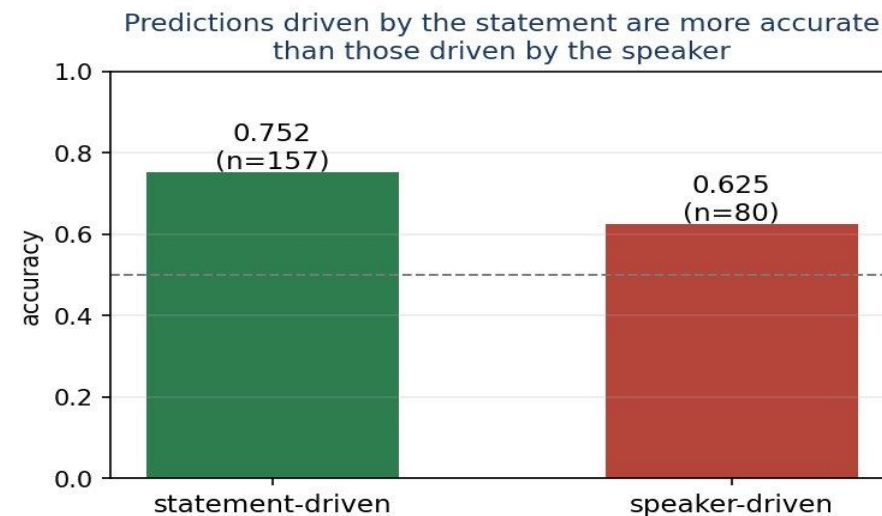
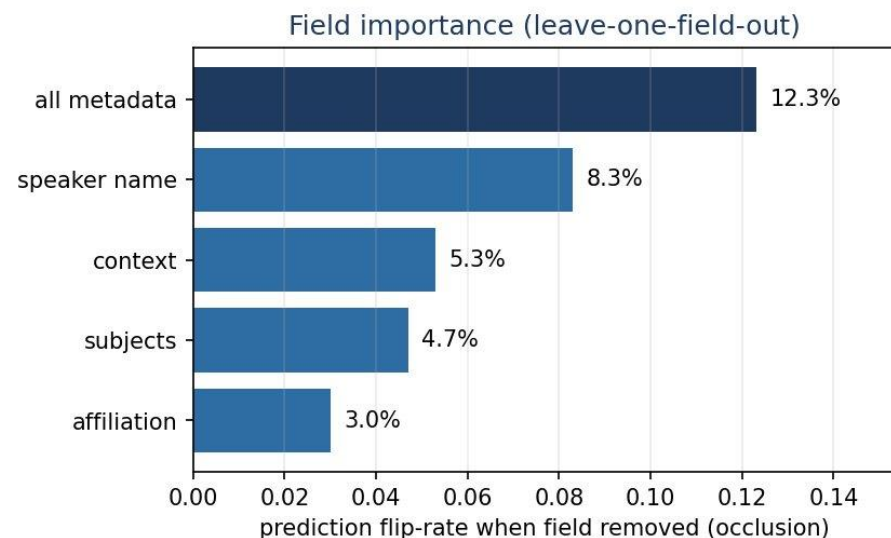
Calibration: The Real Win

- Raw fine-tuned confidence is badly overconfident (SFT pushes probabilities toward 0/1); temperature scaling, fit on validation, corrects it: fine-tuned ECE 0.280 $\rightarrow$ 0.027 on test ($\sim 10\times$ reduction); reliability curve moves onto the diagonal.
- For a tool that acts on confidence (abstain, escalate, flag), trustworthy probabilities are the operationally useful property.



Explainer: Faithful Attribution Beats Plausible Stories

- **explain**(model, points, labels): model-agnostic, either predictor
- **Two layers**
 - 1) *leave-one-field-out occlusion* (faithful by construction)
 - 2) *model's own rationale* (plausible, not necessarily faithful)
- **Findings**
 - removing speaker flips 8.3% of predictions
 - removing all metadata 12.3%
 - rationales only ~46% faithful to the true driver
 - speaker-driven predictions less accurate (statement-driven 0.752 vs speaker-driven 0.625)



Findings & Future Work

- **Findings**

- a low-ceiling task evaluated leakage-free
- fine-tuning gave a significant accuracy gain (+3.3 pts, $p < 0.001$) over zero-shot and halved calibration error
- the explainer exposed a speaker shortcut that hurts and showed self-rationales are unreliable

- **Future work**

- logprob elicitation everywhere
- richer calibration (isotonic)
- ensembling *zero-shot* and *fine-tuned*
- alternative elicitations (score-mode trade-off)